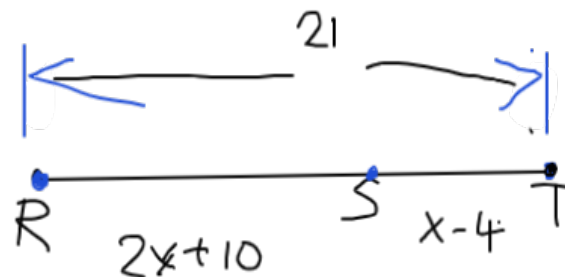


8)

MATHEMATICAL CONNECTIONS Point S is between points R and T on $\overline{RT}$. Use the information to write an equation in terms of x . Then solve the equation and find RS , ST , and RT .

- a. $RS = 2x + 10$ b. $RS = 3x - 16$
 $ST = x - 4$ $ST = 4x - 8$
 $RT = 21$ $RT = 60$



$$(3x-16) + (4x-8) = 60$$

$$7x - 24 = 60$$

$$7x = 84$$

$$x = 12$$

$$(2x+10) + (x-4) = 21$$

$$3x + 6 = 21$$

$$3x = 15$$

$$x = 5$$

a) $RS = 20$ units

$ST = 1$ unit

b) $RS = 20$ units
 $ST = 40$ units

Midpoints and Segment Bisectors

Core Concept

Midpoints and Segment Bisectors

The **midpoint** of a segment is the point that divides the segment into two congruent segments.



M is the midpoint of $\overline{AB}$.

So, $\overline{AM} \cong \overline{MB}$ and $AM = MB$.

A **segment bisector** is a point, ray, line, line segment, or plane that intersects the segment at its midpoint. A midpoint or a segment bisector *bisects* a segment.

Identify the segment bisector of $\overline{PQ}$. Then find PQ .

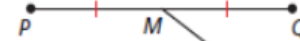
1. $\overline{1\frac{7}{8}}$

2.

$\overline{2\frac{2}{7}}$



$\overrightarrow{CD}$ is a segment bisector of $\overline{AB}$.
So, $\overline{AM} \cong \overline{MB}$ and $AM = MB$.

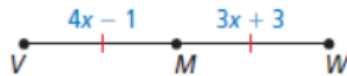


Bisector = $\overrightarrow{MN}$
 $\overline{PQ} = \frac{30}{8}$

Bisector = M
 $\overline{PQ} = \frac{32}{7}$

9)

Point M is the midpoint of $\overline{VW}$. Find the length of $\overline{VM}$.



$$4x - 1 = 3x + 3$$

$$x = 4$$

$$\overline{VM} = 15 \text{ units}$$

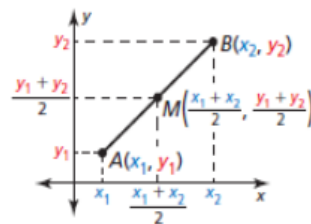
Core Concept

The Midpoint Formula

The coordinates of the midpoint of a segment are the averages of the x -coordinates and of the y -coordinates of the endpoints.

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are points in a coordinate plane, then the midpoint M of $\overline{AB}$ has coordinates

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

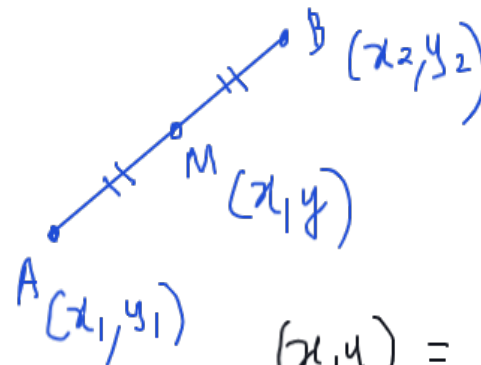


EXAMPLE 3 Using the Midpoint Formula

- The endpoints of $\overline{RS}$ are $R(1, -3)$ and $S(4, 2)$. Find the coordinates of the midpoint M .
- The midpoint of $\overline{JK}$ is $M(2, 1)$. One endpoint is $J(1, 4)$. Find the coordinates of endpoint K .

SOLUTION

- Use the Midpoint Formula.



$$(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

a. Use the Midpoint Formula.

$$M\left(\frac{1+4}{2}, \frac{-3+2}{2}\right) = M\left(\frac{5}{2}, -\frac{1}{2}\right)$$

► The coordinates of the midpoint M are $\left(\frac{5}{2}, -\frac{1}{2}\right)$.

b. Let (x, y) be the coordinates of endpoint K .

Use the Midpoint Formula.

Step 1 Find x .

$$\frac{1+x}{2} = 2$$

$$1+x = 4$$

$$x = 3$$

Step 2 Find y .

$$\frac{4+y}{2} = 1$$

$$4+y = 2$$

$$y = -2$$

► The coordinates of endpoint K are $(3, -2)$.

